MILITARY SPECIFICATION

MANOMETER, SPINAL FLUID, WATER TYPE, DISPOSABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of a sterile disposable spinal fluid manometer suitable for measuring spinal fluid pressure.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

MMM-A-250	Adhesive (Water Resistant) for Sealing Fiberboard Boxes).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

Military

MIL-S-36586	Sterilization Strip Test, Bacterial Spore.

FSC 6515

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)
24028 MANOMETER, SPINAL FLUID, WATER TYPE, DISPOSABLE

(Copies of specifications, standards and drawing required by suppliers in connection with specific procurement functions should be obtained from the procurement activity or as directed by the procurement officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply:

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.
The Pharmacopeia of the United States of America (U.S.P.)

(Application for copies shall be addressed to the Mack Publishing Company, Easton, Pennsylvania 18042.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)
ANSI Z70.1 American Standard Dimensions of Glass and Metal Luer Tapers for Medical Applications.

(Application for copies should be addressed to the American National Standards Institute (ANSI, 1430 Broadway, New York, N.Y. 10018.)

3. REQUIREMENTS

3.1 Material. Tube and stopcock shall be fabricated from plastic material suitable for the purpose intended. Plastic material for tube shall be water clear.

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3.2 Style, design and dimensions. Set shall be a spinal fluid manometer consisting of a capillary tube and a three way stopcock, or a four way stopcock which allows the fluid to be collected in a vertical position. Shall be in accordance with type I or type II of DPSC Drawing 24028. The dimensions shown in the drawing are for reference purposes and are intended to indicate general style and design only.

3.2.1 Capillary tube. The capillary tube shall be calibrated from 3 centimeters (30 mm) to 55.0 cm (550 mm) minimum. The scale shall be marked in 2 millimeter graduations with main graduations every 1 centimeter (10 mm).

3.2.2 Stopcock. Shall be a three way or four way stopcock and shall have one Luer male tip and two Luer female lock connections.

3.2.3 Male and female luer connectors. The male connector shall have a tapered tip to receive a standard hypodermic needle. The tip shall make a mechanically stable, fluid tight connection with a hypodermic needle having a Luer hub with hub cavity in accordance with Figure 2 (female end) of American National Standards Institute (ANSI) Standard Z70.1. The female connector shall have a tapered cavity to receive a standard hypodermic syringe. The cavity of the female connector shall make a mechanically stable, fluid tight connection with a syringe having a Luer tip in accordance with Figure 4 of ANSI Standard Z70.1.

3.3 Fluid tightness of assembly. The assembled capillary tube and stopcock assembly shall pass the fluid test as specified in 4.4.1. When so tested, no point shall show leakage exceeding the formation of a falling drop in 30 seconds.

3.4 Sterility. The fluid paths and end connectors shall be sterile when tested with a biological indicator as specified in 4.4.4.

3.4.1 Certificate of sterility. A certificate of sterility, stating that samples from each sterilizer lot have been tested and found to be sterile, shall be submitted to the procuring activity. The certificate of sterility shall include:

a. Item identification.
b. Federal stock number.
c. Contract or purchase order number.
d. Sterility lot or control number.

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3.4.1 Certificate of sterility. (cont'd)

e. Date of sterilization.
f. Quantity of product in sterilizer lot.
g. Type of sterilization (gas, radiation, heat, steam, etc.).
h. Form of biological indicator (i.e., inoculated paper spore strip or carrier, inoculated product, or inoculated simulated product).
i. Test organism(s) and population density used.
j. Test medium, incubation temperature, incubation time, and dates of tests.
k. Number of test units.
l. Statement that samples comply with sterility tests.
m. Signature of head of laboratory or a designated qualified alternate.

When inspection is made at destination, the certificate of sterility shall be submitted by the contractor to the Directorate of Medical Materiel, ATTN: DPSC-ATQ, Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pa., 19101, and a copy forwarded with each shipment to the consignee. When inspection is made at source, certificate of sterility shall be furnished to the cognizant Government Quality Assurance Representative for submission to the procuring activity.

3.4.2 Sterility lot. A sterility lot is that single quantity of product subjected to the same manufacturing operation and simultaneously sterilized in the same sterilizer chamber. Each unit package shall be clearly marked with a sterility lot (control) number and date of sterilization as specified in 5.3.1. Each sterility lot shall be traceable to the pertinent manufacturing lot(s).

3.5 Safety (non-toxicity). The tubes shall be safe when tested for safety as specified in 4.4.2. All equipment shall be safe.

3.6 Freedom from pyrogens. The fluid path and end connectors shall be pyrogen free when tested as specified in 4.4.3.

3.7 Certificates of freedom from pyrogens and safety (non-toxicity)test. A certificate listing all lot or control numbers shall state that representative samples from each lot or control have been tested in accordance with methods described herein, and found to be free from pyrogens and have passed the safety (non-toxicity) test. This certificate shall include the

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specific tests performed and the results. These certificates shall be submitted by the contractor in the manner as specified in 3.4.1.

3.8 Manufacturer's record. The manufacturer shall place a code or lot number on the paper label which can be verified against the manufacturer's records of all tests made on nontoxic, sterile, pyrogen-free sets. These records shall be made available for inspection by any duly authorized Government representative. No shipment shall be made until all tests have been completed, verified, and recorded in the permanent records of the manufacturer.

3.9 Compatibility. The material of which sets are manufactured shall not react with, or be acted upon by, the liquids with which they will be in contact during normal use and operation, nor shall there by any material incorporated in the set which will cause any harmful or deleterious effect when the set is used for the purpose intended.

3.10 Workmanship. Workmanship shall be first class in every respect. Sets shall be free from defects which detract from their appearance or may impair their serviceability.

3.11 Instructions. Instructions for use shall be furnished with each individual package.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the plastics, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. The unit of product for examination purposes shall be one manometer. Non-compliance with any one of the special requirements shall be cause for rejection.

Table I. Sampling Criteria for Examination

Type Defect	Inspection level	AQL (percent defective)
Critical	None defined	
Major	II	1.0
Minor	II	2.5

4.2.2 For tests.

4.2.2.1 For sterility test. Sampling for sterility test shall be in accordance with the U.S.P. for parenteral devices and as specified in 4.4.4.

4.2.2.2 For pyrogen tests. Sampling for pyrogen tests shall be accomplished by selecting samples which are to be representative of each lot (not less than two manometers) of material used in the fabrication of the manometers.

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4.2.2.3 For other tests. Sampling for other tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. The unit of product for testing purposes shall be one manometer. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics. Defective manometers shall be rejected.

Table II. Sampling Criteria for Tests

Component	Characteristic	Reqmt.	Test procedure	Unit of product	Insp. level	AQL (percent defective)
Manometer	Fluid tightness	3.3	4.4.1	1 Set	S-1	*

*Acceptance number shall be zero.

4.3 Examination. The sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in Table I and III. Examination, however, shall not be restricted to defects listed in Table III. Any units in the sample containing one or more defects shall be rejected and if the number of defective units in any sample exceeds the acceptance number for the sample, the lot represented by the sample shall be rejected.

Table III. Classification of defects

Categories	Defects
Critical	None defined.
Major	
101	Manometer not calibrated in 2 mm graduations.
102	Manometer not calibrated between 3 and 55 cm.
103	Manometer not free of dust, dirt or other foreign matter.
104	Immediate package does not open readily.
105	Immediate package not free of punctures, tears or open seals.
106	Three way stopcock does not operate and flow freely.
107	Set does not contain one tube and one stopcock.
108	Manometer tube not fabricated of water clear plastic.

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Table III. Classification of defects (cont'd.)

Categories	Defects
Minor	
201	Luer tips not free of flash.
202	Set not free of defects that detract from appearance but do not affect serviceability.
203	Packaging, packing and marking not as specified.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Fluid tightness of assembly. The assembled capillary tube and three way stopcock shall be tested for fluid tightness as follows: The capillary tube shall be fitted to one of the stopcock hubs. A gage, as shown on Figure 8 of ANSI Standard Z70.1 (except that gage shall have a fluid path) shall be fitted to the male Luer slip of the stopcock. The other hub of the stopcock shall be attached by means of a plug gage as shown in Figure 5 of ANSI Standard (except that the gage shall have a fluid path) to a water pressure source having a pressure gage. Water shall be run through the assembly to fill any cavity spaces. The lumen of the gage on the male tip shall be sealed by a piece of solid rubber and the distal end of the capillary tube shall be capped. A pressure of 8 p.s.i. shall be applied to the apparatus for 30 seconds. This test shall be performed for both female Luer hubs, male Luer slip, capillary tube in stopcock as well as joints between the spigot and stopcock body.

4.4.2 Safety (non-toxicity). The set shall be tested for safety in accordance with the Pharmacopeia of the United States (U.S.P.) for infusion and transfusion assemblies.

4.4.3 Freedom from pyrogens test. The pyrogen test shall be performed as described in the U.S.P. for infusion and transfusion assemblies.

4.4.4 Sterility test. The fluid paths and end connectors shall be tested for sterility in accordance with the method specified in the USP for infusion and transfusion assemblies using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates from product intended for distribution. The incubation medium and minimum time in each case shall be soybean casein digest for 7 days for inoculated product, inoculated simulated product, or paper spore strip (carrier). The incubation temperature shall be as indicated

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in paragraph 4.4.4.1. The minimum number of test units, regardless of lot size, shall be 10, except for inoculated carrier which shall be 20 for Type I Sterilization and 30 for Type II Sterilization including inoculated carrier and product test units. The inspection level shall be as indicated in Table II, Sampling Criteria for Tests.

4.4.4.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism	Incubation Temperature
Steam	B. stearothermophilus	55° - 65°C
Dry Heat	B. stearothermophilus, or	55° - 65°C
	B. subtilis var.niger (or globigii)	30° - 35°C
Gas	B. subtilis var. niger (or globigii)	30° - 35°C
Radiation	B. pumilus	30° - 35°C

The number of viable cells shall be sufficient to provide assurance that the product has been sterilized.

4.4.4.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used shall be in accordance with MIL-S-36586. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.4.4 Sterility test and 4.4.4.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.4.4.1.
NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP.

4.4.4.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer, and approval obtained before employment of the alternate method on a government contract.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified see (6.2).

5.1.1 Unit of issue. One package, contining, 10 manometers, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in the following table shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	5 units	20 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Individual package. Each manometer shall be packaged in a sealed envelope designed to maintain sterility of the item until manually opened.

5.1.3.2 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.3.3 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flap with adhesive conforming to MMM-A-250 or metal stitches is permitted.

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5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual package. Each package shall be marked with the item identification, manufacturer's name and address and the lot or control number. In addition the word "STERILE" and the legend "STERILITY GUARANTEED UNLESS PACKAGE IS DAMAGED OR OPENED" or similar statement.

5.3.2 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is

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fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. Marking shall include the lot or control number. In addition, the method of sterilization (G-Gas, R-Radiation, C-Cold, S-Steam, D-Dry Heat Sterilization) shall be shown following the level of packaging and date of packaging. Example: "A-2/73/G". Note: Final sterilization shall be on or after indicated packaging date.

5.3.3 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Marking shall include the lot or control number and the method of sterilization as specified in 5.3.2.

6. NOTES

6.1 Intended use. The manometers are intended for use in measuring the pressure of the spinal fluid, allowing the fluid to be collected in a vertical position.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification is intended to cover the following item appearing in the Federal Supply Catalog:

Federal Stock No.	Item Identification
6515-782-2620	MANOMETER, SPINAL FLUID, Water Type, Disposable, 10s

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

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6.4 Supersession data. This specification includes requirements of Defense Medical Purchase Description No. 5, dated 30 August 1968: "for FSN 6515-782-2620 - MANOMETER, SPINAL FLUID, Water Type, Disposable, 10s."

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review activities:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:

Defense Supply Agency - DM

Project No. 6515-1711

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. *DoD contractors, government activities,* or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. *If there are additional papers, attach to form and place both in an* envelope addressed to preparing activity.

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MIL-M-36958

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

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